TILEncore-Gx36 Card User's Guide

Featuring the TILE-Gx36 Processor

Release 1.4
Doc. No. UG416
September 2015
EZchip Semiconductor, Inc.

Publishing Information:

Document Number	UG416
Release	1.4
Date	9/14/15

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Contents

CHAPTER 1 OVERVIEW

Thank you for purchasing the TILEncore-Gx36™ Card featuring EZchip's revolutionary multicore Tile Processor™.

Note: In this document, "TILEncore-Gx36" is used to refer to various versions of the card. Models are offered with 4x SFP+ ports.

1.1 In this Release

The following items, shown in Figure 1-1, are included in this shipment:

- TILEncore-Gx36, featuring the TILE-Gx36™ processor. The card comes installed with the following version of the processor:
  - TILE-Gx8036™ (Series 8000)
- Standard USB cable micro-to-A (not shown).
- RS232 cable
- Printed documentation, including the *TILEncore-Gx36 Card User's Guide* (UG416) (*this document*).

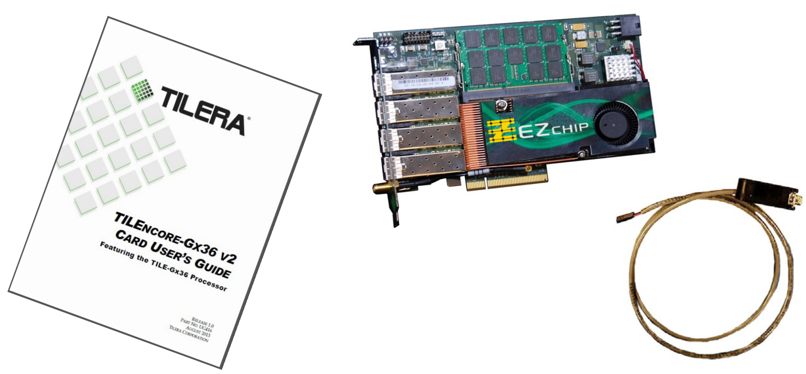

Figure 1-1: What is in the Box

Optional and Customer-Supplied Parts

- Optional cables. Customers must provide appropriate interface cables as needed. Refer to Section 4.3 *"Connecting Cables"* on page 21.

Interface Module Requirements

If high-speed Ethernet networking is required, customer-supplied SFP or SFP+ modules must be installed into the TILEncore-Gx board. Between 1 to 4 modules can be inserted, which can be a mix of SFP and SFP+ modules. Note that when the network interfaces are configured, the speed (1G/10G) must be specified to match the installed cartridge.

The modules that have been qualified for use are listed in Table 5-4, "Qualified SFP Optical and Copper Modules," on page 29 and Table 5-5, "Qualified SFP+ Optical Modules," on page 30.

1.2 About this Guide

The *TILEncore-Gx36 Card User's Guide* describes the installation, setup, and features of EZchip's TILE-Gx36™ processor. This book includes the following chapters:

- Chapter 1: Overview (*this chapter*) describes the contents of this manual.
- Chapter 2: About the TILEncore-Gx36 identifies the intended user for the TILEncore-Gx36 and the Card's features and supported interfaces.
- Chapter 3: System and Board Requirements specifies the power and system requirements, including the hardware and software requirements that should be followed in order to operate the Card properly.
- Chapter 4: Installing the TILEncore-Gx36 provides pre-installation guidelines and proper handing instructions.
- Chapter 5: Functional Description describes some of the card's features and subsystems, specifically the card resets, LEDs, interrupts, and boot options.
- Chapter 6: Running Programs on the TILEncore-Gx36 provides instructions for booting the card and running a sample program.
- Chapter 7: Verifying that the Card Operates Properly describes how to test the card to verify that it has been installed properly.
- Appendix A: Specifications provides electrical, mechanical, and environmental specifications for the Card.
- Appendix B: Safety and Regulatory Compliance lists the safety and compliance information for the TILEncore-Gx36.
- Appendix C: Troubleshooting
- Glossary provides definitions for terms used in this document.

An Index is also included.

1.2.1 Intended Audience

This book is intended primarily for hardware developers who want to use the TILEncore-Gx36 to jump start whole system designs, designers evaluating the TILE-Gx36™ processor for use in their boards, and individuals looking for a system solution. The TILEncore-Gx36 is offered separately from but in conjunction with the Multicore Development Environment™ (MDE), accompanying tools, software libraries, and software documentation. Contact your support engineer or sales representative for more information about the MDE.

1.2.2 Notation Conventions

Table 1-1 lists document notation conventions.

Table 1-1. Notation Conventions

Example	Description
Close command (File menu)	Titles in reference sections indicate the location of an item within the Graphical User Interface (GUI) for the MDE (for example, the Close command appears on the File menu).
Write In Progress (`WIP`) bit	`Courier` text indicates the names of: • A bit or bitfield, for example the `CHANNEL` bit • A special purpose register (SPR), for example the `RSH_COORD` SPR • Code sample • Application, for example the `tile-monitor` application • Command, for example the `link` command • State, for example the `IDLE` state
`RSHIM MMIO registers`	Blue and underlined text in `Courier` font indicates these are hypertext links to HTML files associated with the interface, or links to a specific web site.
Appendix B: Safety and Regulatory Compliance	Blue text in text font (Palatino) indicates a (cross-reference) Hypertext link to text elsewhere in the manual. If you are reading a soft-copy of this manual, you can click on the link to jump directly to the referenced section.
{this | that}	Alternative required items in syntax descriptions appear within curly brackets and separated by vertical bars; read the example as this or that. One or the other is required.
[this | that]	Optional items in syntax descriptions appear within brackets and separated by vertical bars; read the example as an optional this or that.
[this,…]	Optional item lists in syntax descriptions appear within brackets delimited by commas and terminated with an ellipse; read the example as an optional comma-separated list of this.
.SECTION	Commands, directives, keywords, and feature names are in text with Arial font.
<filename>	Non-keyword placeholders appear in text in italic style format and within brackets.
`www.ezchip.com`	Worldwide web addresses, directory names, code samples all share a `courier` font.
	**Note:** For correct operation, ... A Note provides supplementary information on a related topic. In the online version of this book, the word **Note** appears instead of this symbol.
	**Caution:** Incorrect device operation might result if ... **Caution:** Device damage might result if ... A Caution identifies conditions or inappropriate usage of the product that could lead to undesirable results or product damage. In the online version of this book, the word **Caution** appears instead of this symbol.

1.3 Technical and Customer Support

This section describes how to obtain additional information.

1.3.1 Customer Support

You can reach EZchip Semiconductor, Inc. Customer Support in either of two ways:

- Visit the EZchip® Web site at

  `http://www.ezchip.com/`

- E-mail questions to

  `supportMC@ezchip.com`

1.3.2 Product Information

You can obtain additional product information from the EZchip Semiconductor, Inc. Web site, from the product CD-ROM, or from EZchip printed publications.

Access the EZchip Web site online at:

`http://www.ezchip.com/`

1.3.3 Related Documents

Additional documentation is available upon request. Most of the documentation listed below is included with EZchip's MDE.

1.3.3.1 Hardware Documents

Hardware Design

- *TILE-Gx36 Processor Data Sheet* (DS400)
- *Hardware Design Guidelines for TILE-Gx Processors* (WP006)

Architecture

- *Architectural Overview for the TILE-Gx Series* (UG130)
- *TILE-Gx Instruction Set Architecture* (UG401)
- *Tile Processor and I/O Device Guide for the TILE-Gx Family of Processors* (UG404)

Boards

- *TILEncore-Gx36 Card User's Guide* (UG416) (*this manual*)
- *Installation Sheet for: TILEncore-Gx36* (IS104)
- *Board Test Kit Guide for TILE-Gx Processors* (UG422)

1.3.3.2 Software Documents

The EZchip MDE Document Index provides links to all the MDE documentation, including HTML and PDF files and additional documentation:

`${TILERA_ROOT}/doc/index.html`

Note: `${TILERA_ROOT}` is the root of the EZchip MDE install directory.

1.4 MDE Documentation

Note: `TILERA_ROOT` is an environment variable that points to the root of the MDE installation directory. As such it is a convenient shorthand in the documentation. It is required by many EZchip tools and examples. We recommend that you use the `tile-env` script described in *Getting Started with the Multicore Development Environment* (UG504) to set it correctly.

EZchip MDE Document Index

The EZchip MDE Document Index provides links to online copies of the MDE documentation, in both HTML and PDF formats:

```
$TILERA_ROOT/doc/index.html
```

Location of PDF Files in the Kit

The PDF versions of MDE documents can be found at this location:

```
$TILERA_ROOT/doc/pdf/
```

Man Pages and Info Pages

To view a man page for an x86 tool, run:

```
$ tile-man toolname
```

To view a man page for a Tile tool, run this on the x86 host:

```
$ tile-man toolname
```

IDE Online Help

The IDE has its own IDE Online Documentation, which you can access when you use the IDE.

To get to the online help:

- Start `tile-eclipse`.

Select the menu command Help → Help Contents.

- When the online help browser is displayed, select the document IDE Online Documentation in the left navigation pane, if it is not already selected.

Chapter 2 About the TILEncore-Gx36

2.1 Product Overview

The TILEncore-Gx36™ PCIe card is a PCIe endpoint that enables customers to develop and test software applications on the TILE-Gx36™ processor. Physically, it is a full-height, ½-length, single slot PCI Express form factor with an 8-lane edge connector. Up to four network interfaces are provided on the PCIe rear plate. Each of these interfaces is an SFP cage that can support either 10Gb (SFP+) or 1Gb Ethernet (SFP).

TILEncore-Gx36 is a modular card with a processor and SODIMM memory.

2.2 Purpose

The TILEncore-Gx36 Card serves three purposes: as an early development vehicle, as evaluation hardware for the multicore processor, and for integration into a "black box" system for developers who do not want to build their own hardware. Due to the wide variety of usage models, no software applications accompany the TILEncore-Gx36, but they can be obtained separately (see "Customer Support" on page 4).

2.3 Features

- TILE-Gx™ processor capable of running from 1.0 GHz to 1.2 GHz.
- Two modular DDR3 (ECC SODIMM) memory interfaces.
- Various interface options:
  - Four SFP/SFP+ connectors.
  - RS-232 (UART) port accessed via a 3-pin header (the header pins are the same as on other EZchip® PCIe cards).
  - Standard USB accessed via micro-to-A cable
- PCI Express standard, gold-fingered edge connector to Host, which uses an eight-lane physical connector.
- User-selectable boot options, including SPI Flash ROM and host boot via PCIe.

  The on-board serial Boot ROM(s) are shipped with the `prebooter` (to configure the PCIe link). Initial booting is expected to be via the PCIe.
- Temperature monitoring of the environment and the TILE-Gx core.
- Power monitoring via software.
- The DDR voltage is jumper-selectable only (1.35V or 1.5V). Core voltage is selectable via VIDs.

2.3.1 Design and Layout

The TILEncore-Gx36 is comprised of two primary sections:

- The TILE-Gx processor (and its memory subsystems)
- The interface subsystem with SFP/SFP+ connectors

Figure 2-1 and Figure 2-2 depicts the major card components and associations.

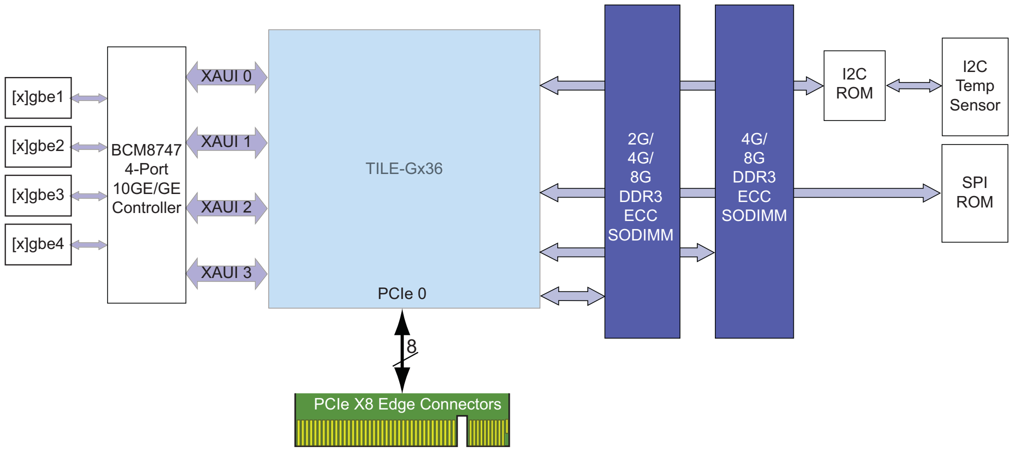

Figure 2-1: TILEncore-Gx36 Block Diagram

Figure 2-2 shows a top view of the board and identifies the major components. For locations of the LEDs, refer to Figure 7-1.

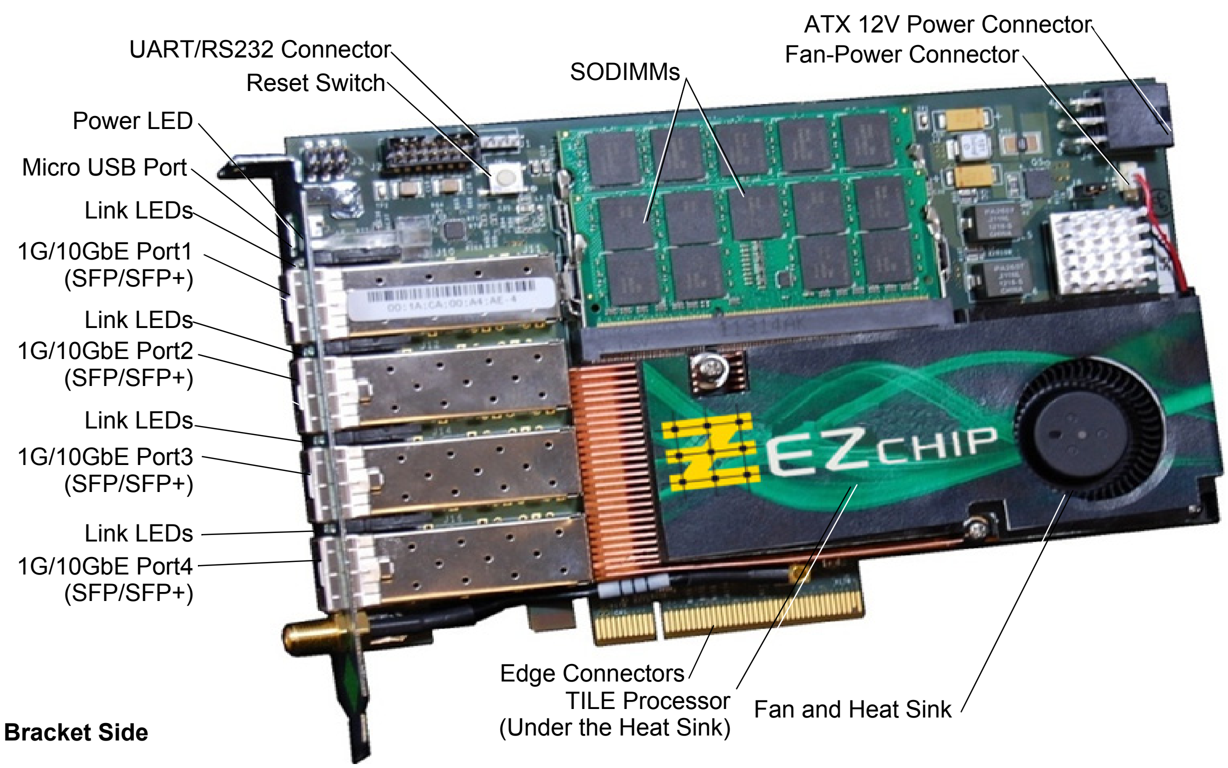

Figure 2-2: TILEncore-Gx36 with SFP/SFP+ Interfaces

Figure 2-3 shows the bottom view of the main board and the location of the DIP switches.

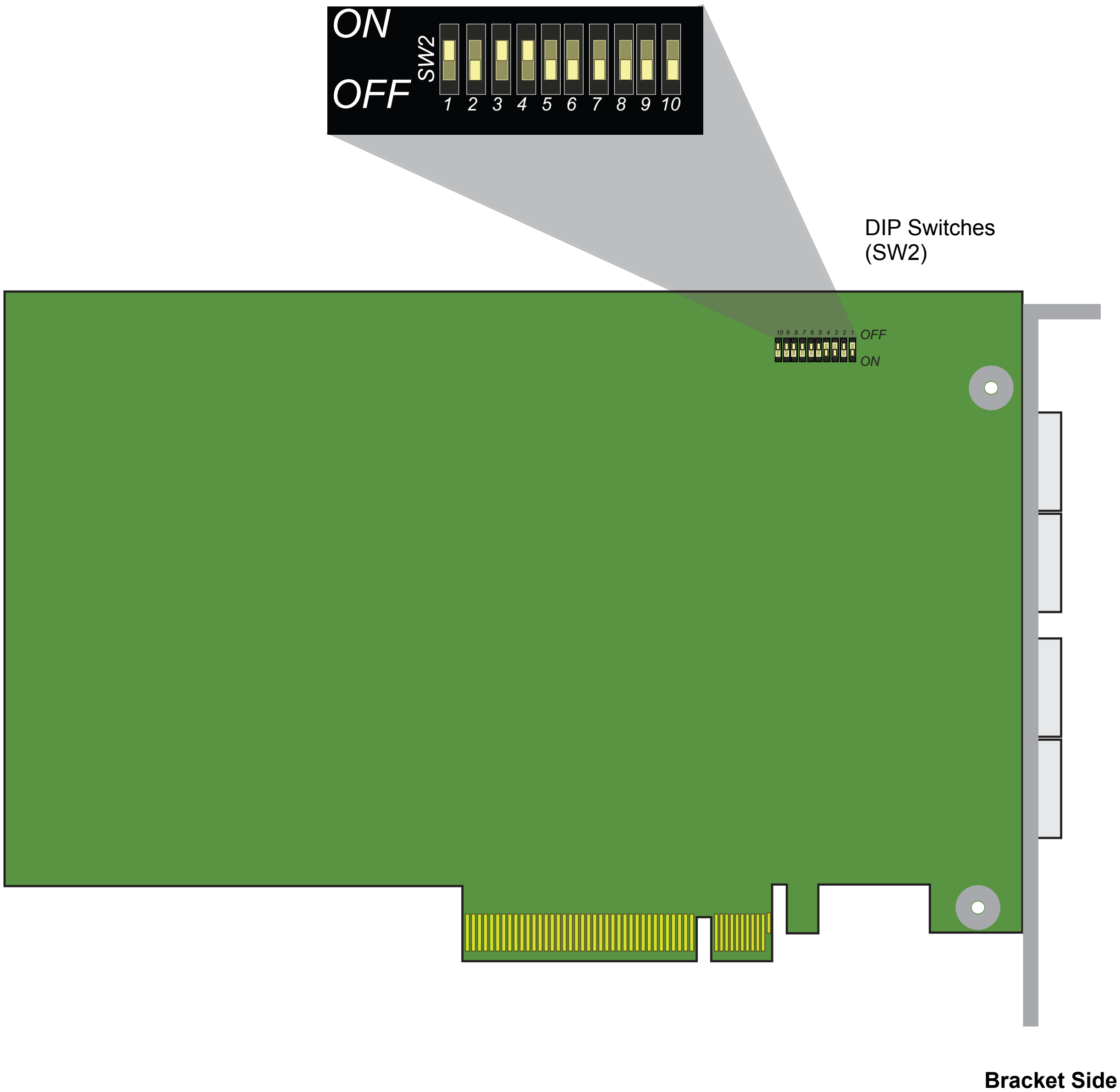

Figure 2-3: TILEncore-Gx36 — Bottom View

CHAPTER 3 SYSTEM AND BOARD REQUIREMENTS

3.1 System Requirements

This section lists minimal requirements for installing and using the TILEncore-Gx36™.

3.1.1 Hardware Requirements

Minimum system hardware requirements are described in the sections that follow.

3.1.1.1 Host System

EZchip® has verified that the TILEncore-Gx36 card works many systems. For a complete list of qualified systems, refer to "PCIe Host Requirements" in the *MDE Patch Notes* (UG718) and *MDE Release Notes* (UG708).

3.1.1.2 Mechanical Requirements

As a half-length, full-height (HF) PCIe add-in card, the TILEncore-Gx36 installs in a standard x8 or x16 PCIe slot and conforms to PCIe clearance specifications for top and bottom component heights as well as specified keepouts.

3.1.1.3 Mother Board Requirements

The TILEncore-Gx36 supports a x8 PCIe interface and requires a x8 PCIe connector. Higher bandwidth connectors are backward compatible; therefore, a x16 connector works. From an electrical standpoint, the TILEncore-Gx36 trains up to Gen 2 x8 (maximum).

3.1.1.4 Air Flow

For boards installed in host systems, adhere to the following recommended operating conditions:

CAUTION!

Airflow and ambient temperature requirements:

– 100 lfm @ 40°C with no adjacent card

– 400 lfm @ 40°C with adjacent card

Note: Do not leave the PC cover open as it can disrupt air flow inside the PC. Air circulation is best with the cover closed.

3.1.2 Software Requirements

Minimum system software requirements are described in the sections that follow.

3.1.2.1 Operating System Requirements

In order for you to operate the TILEncore-Gx36, you need the following software installed on your system:

- Operating System software should be Multicore Development Environment™ (MDE) Release 4.1.7 or later.

  Refer to the *MDE Release Notes* (UG708) for specific requirements.
- Red Hat Enterprise Linux software. The MDE has been qualified on a host system running the appropriate releases of RHEL and CentOS, as specified in the MDE Release Notes (UG708) and/or MDE Patch Notes (UG718).
- EZchip includes pre-built drivers for Red Hat Enterprise Linux 5, CentOS 5, Red Hat Enterprise Linux 6, and CentOS 6. The driver source is provided; other kernels need to compile their own drivers. (EZchip provides build scripts, for example: `$TILERA_ROOT/lib/modules/tilepci-compile`). Refer to MDE Release Notes (UG708) and/or MDE Patch Notes (UG718) for a list of these pre-built drivers.

  The *Linux Red Hat Installation Guide* can be found at:

  `http://docs.redhat.com/docs/en-US/Red_Hat_Enterprise_Linux/5/html/Installation_Guide/index.html/`

For instructions on configuring drivers, refer to *"I/O Devices and Drivers"* in the *MDE System Programmer's Guide*. For more information, contact `supportMC@ezchip.com`.

3.1.2.2 Development Software Requirements

The following software is required:

EZchip Multicore Development Environment (MDE). You must use a current copy of EZchip's MDE— at a minimum 4.1.7 revision level. This developer's kit provides drivers and runtime code required to use the TILEncore-Gx36.

3.2 Power Requirements

The TILEncore-Gx36 can get power from either the power cable (ATX power cable) or the edge connectors.

- +12V from the host PC power supply. The TILEncore-Gx36 gets the +12V exclusively from the PCI Express 150W-ATX (6 pin) Power Connector located on the top edge of the card (see Figure 4-3).
- The TILEncore-Gx36 can get power from the PCIe "edge connectors".

3.2.1 ATX Cable

3.2.1.1 Specifications

- +12V: Regulation tolerance = +/-5% (11.4V - 12.6V)
- Output ripple = 120 mV pp
- +5V: This is a NO-CONNECT on TILEncore-Gx36.
- The PCIe card does not tolerate "sag" or transients outside of these tolerances.

For power consumption guidelines, refer to Section A.1 *"Electrical Specifications"* on page 47.

Note: The board must consume no more than:
65 watts if connected using PCIe fingers
75 watts if connected using ATX connector

Cumulative power load on the power supply must be considered when installing one or more TILEncore-Gx card(s) in a host system. Always consider the power draw required by other PCIe cards installed in the system, including the graphics card (GPU) as well.

CHAPTER 4 INSTALLING THE TILENCORE-GX36

4.1 Pre-Installation Guidelines

This section describes steps to take when installing the TILEncore-Gx36™ card.

4.1.1 Unpacking the TILEncore-Gx36 Card

Before unpacking your TILEncore-Gx36, inspect the shipping carton for damage. If the shipping container appears damaged, notify the shipper and EZchip® immediately.

- **When you unpack your TILEncore-Gx36 card:**

Carefully remove packing material. Retain the original packing in case the unit must be returned for servicing or being shipped.

4.1.2 Handling the TILEncore-Gx36 Card

4.1.2.1 Preventing Electrostatic Discharge (ESD)

In order to prevent ESD during shipment and storage, the TILEncore-Gx36 is placed in a static shielding bag, seated in an anti-static foam cushion, and placed in a box. This protection prevents vibration between the card and the box and avoids possible buildup of static electricity and ESD damage.

Before opening the box place it on an ESD-protected work area, preferably on a static dissipative table top. Workers should further protect the card by wearing a wrist strap that is connected to a ground point. The wrist strap is the primary means of controlling static charges on personnel. A non-conductive flooring is also recommended as walking across a conductive surface causes static electricity to build up.

When the card is not inserted in a PC, it should be kept in the protective bag in which it was shipped.

4.1.2.2 Preventing Damage to the TILEncore-Gx36 Card

Handling

When handling the TILEncore-Gx36 always hold it by the edges, not by any components, connectors or fixtures installed on the card. Do not flex the card as this can damage the solder connections.

4.1.3 Preparing for Installation or Removal

Tools Needed

Table 4-1 lists the tools needed for removing and installing the TILEncore-Gx36 card in your PC.

Table 4-1. Removal and Replacement Tools

Task	Tools
Removing/installing the TILEncore-Gx36	• Phillips screwdriver (no. 2), depending on PCI bracket style • Antistatic (ESD) wrist strap
Attaching/disconnecting the UART cable connector from the PC serial port	• Slotted (flathead) screwdriver (no. 1) • Antistatic (ESD) wrist strap

Note: It is important to use the correct size screwdriver in order to avoid damaging the screw heads.

Use caution when working inside the computer chassis as touching or bumping the card's rotating blower module can damage it.

4.2 TILEncore-Gx36 Card Installation

This section provides installation and removal guidelines only.

Note: Some motherboards have restricted slot usage. Refer to your motherboard documentation for this information.

■ Before removing or installing any cards, follow these steps:

1. Assemble the necessary tools.
2. For a standard card installation, first shut down the host system. Ensure that all power is disconnected from the PC in which it is installed.
3. Attach an antistatic wrist guard strap to a ground point.
4. Read the instructions in "Safety and Regulatory Compliance" on page 49 that describe how to handle the TILEncore-Gx36 card.

■ Determine how to power your TILEncore-Gx36 card

Before you install a TILEncore-Gx36 card, you should decide how you want to power your card. You can:

- Power your card via the ATX cable and disable edge connectors
- Power your card via edge connectors, enable edge connectors, and leave the ATX cable unconnected
- Power your card by *either* the ATX cable *or* edge connectors by enabling edge connectors but also connecting the ATX cable

Powering your Card via the ATX Cable

1. Disable edge connector power by setting the DIP SW2 position 9 **ON** (see Table 4-5, “DIP Switch SW2 Settings,” on page 22).

> Note if you have disabled the edge connectors and do not have the ATX cable connected, you cannot bring up the card.

2. Insert the card in any slot in the host system and connect the ATX cable as described in “Install a TILEncore-Gx36 card” on page 17.

Powering your Card via the Edge Connectors

1. Enable edge connector power by setting the DIP SW2 position 9 **OFF** (see Table 4-5, “DIP Switch SW2 Settings,” on page 22).
2. Insert the card in a 75W slot in the host system but *do not* connect the ATX cable.

> If your card is inserted in a slot that does not support 75W and set SW2 to derive power from the edge connectors, a warning appears on the console at startup.
>
> In these cases, you should either connect an ATX cable or use a slot that supports 75W to ensure that power to the card is at the appropriate level.

All the power to the card comes from the edge connectors.

Powering your Card via either the Edge Connectors or ATX Cable (Not Recommended)

1. Enable edge connector power by setting the DIP SW2 position 9 **OFF** (see Table 4-5, “DIP Switch SW2 Settings,” on page 22).
2. Insert the card in a 75W slot and connect the ATX cable as described in “Install a TILEncore-Gx36 card” on page 17.

   When you power up the host system the card is also be powered up as the system negotiates power for you.

Install a TILEncore-Gx36 card

Follow the steps described below to install the hardware and software:

1. Ensure that the TILEncore-Gx36 card DIP switch and jumpers are set correctly. Note that the TILEncore-Gx36 card has one DIP switch labeled SW2, located on the reverse side of the card (see Figure 4-1).

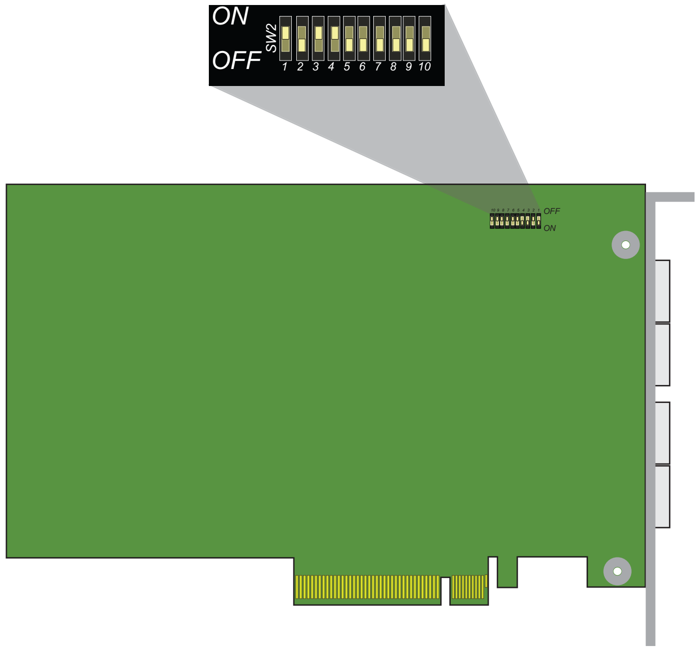

Figure 4-1: TILEncore-Gx36 — Bottom View

Please note that Figure 4-2 shows a graphical representation of the DIP switches with the bracket positioned on the left and gold fingers (also referred to as edge connectors) on the top.

Note: Position-9 of the SW2 switch on the TILEncore-Gx36 is set to the **OFF** position (default) to enable power from the edge connectors.

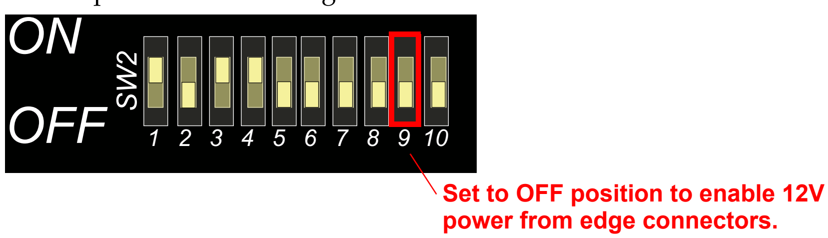

Figure 4-2: Default DIP Switch Settings

2.Power down the host system.

3.TILEncore-Gx36 can be powered from either the PCIe slot (if installed in a 75W- capable slot) or the external 12V ATX power connector.

- If you are powering the card from the PCIe edge connectors, firmly insert the TILEncore-Gx36 into the PCIe x8/x16 slot (Figure 4-3). Position-9 of SW2 should be OFF.

— OR —

- If you are powering the card from the onboard 12V ATX power connector, connect the 6-pin ATX 12V cable from the host power supply to the onboard ATX 12V power connector (Figure 4-3). Position-9 of SW2 should be ON.

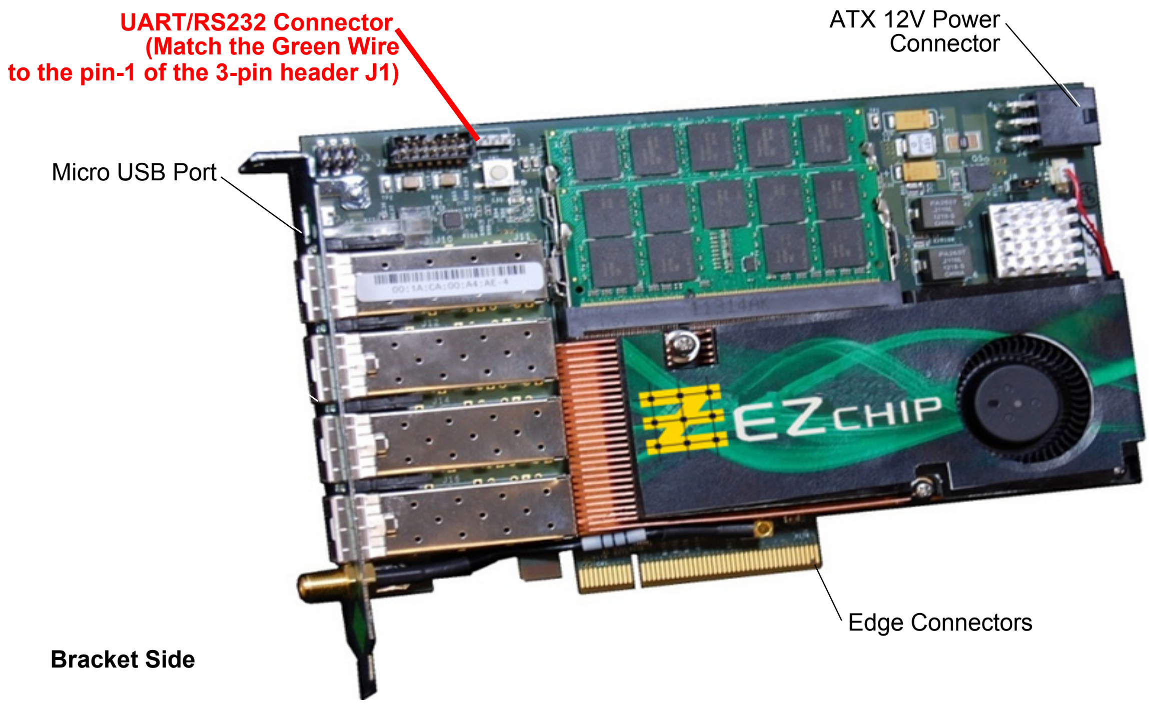

Figure 4-3: TILEncore-Gx36 with SFP/SFP+ Interfaces

4.Fasten the card appropriately for your system, either by snapping the fastening lever or installing the PCI bracket screw securely.

5.Optional: Connect one end of the RS232 cable to the UART/RS232 connector on the card with the green wire matching to pin-1 of the 3-pin header J1, as indicated in Figure 4-3.[1]

6.Connect the free end of the RS232 cable to an available serial port (COM) on your host system. You might need to pass the cable through an opening in the rear of the system.

7.Replace the cover on the system case.

8.Connect the AC power cord and power up the system.

9.Optional: Connect the USB micro-to-A cable, as shown in Figure 4-4. Connect the other end to the Host PC in an available USB port.

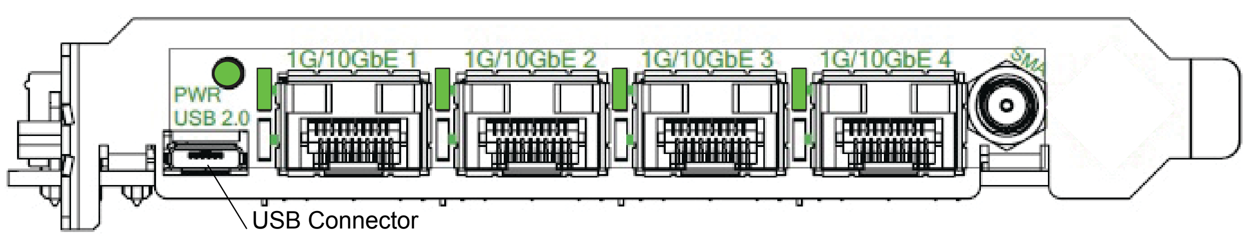

Figure 4-4: Bracket showing the USB Connector

1. Note that there is a "1"mark printed next to pin-1 on the UART/RS232 connector.

10.Install the latest MDE release, as described in *Getting Started with the Multicore Development Environment* (UG504), or the *MDE Release Note*s (UG708) supplied with the MDE. After you have completed the MDE installation process, the PDF versions of MDE documents can be found in: `$TILERA_ROOT/doc/pdf/`.

11.Build the host PCIe driver by running the `$TILERA_ROOT/lib/modules/tilepci-compile` script to build the x86 PCIe driver for your OS version. Refer to *"Building and Installing the x86 Drivers"* as described in the *Getting Started with the Multicore Development Environment* (UG504).

12.Login as `root` and run the driver installation script:

```
# $TILERA_ROOT/bin/install-drivers /dev/ttyS0
```

Without performing this step, you can only call `tile-console` with an explicit serial device name (i.e. `tile-console --serial /dev/ttySx`).

Note: If the card is installed properly, both the PCIe driver and the MDE are installed on the host properly. Open up two terminals and run `tile-console` in one and `tile-monitor` in the other. On the first terminal, type the following command:

`tile-console --dev gxpci0`	To connect to the TILEncore-Gx36 card and display the tile-console.

On the second terminal, type the following command:

`tile-monitor --dev gxpci0`	To connect to the TILEncore-Gx36 card via the tile-monitor.

You should see the standard Tile Linux boot messages in the `tile-console` window. Refer to the *Getting Started with the Multicore Development Environment* (UG504) for more information.

■ Remove a TILEncore-Gx36 card:

1. Remove the PC's cover and place it aside.

   In a hot swap procedure, the host system is still be powered-on and running. Begin by deactivating the slot in which the TILEncore-Gx36 is installed. (Refer to the documentation provided with the PC chassis and/or the operating system.) Note that the green "RUN" LED on the TILEncore-Gx36 goes out, indicating that the board has been powered down and is safe to be removed. Refer to Figure 7-1 for LED locations.

2. Disconnect the micro-USB cable if one is installed.
3. Disconnect the UART cable and the ATX peripheral power cable.
4. Remove any SFPs prior to removing the card.
5. Loosen screws or release the lever holding the card in place.

   Depending on the PC you are using, the card is held in place by either a tool-less fastening lever or a screw. You might need to use a Phillips screwdriver (no. 2) to loosen the screw securing the top of the TILEncore-Gx36 card front bracket.

6. Disengage the card from the chassis, being careful to lift the card up and away from the edge connectors without twisting or flexing the card.

CAUTION! Clearance is limited between card components and the edges of the front panels of adjacent boards. Use care to avoid damaging components as you slide the Card out.

7. Remove the card from the host system and place it in an antistatic bag.

4.3 Connecting Cables

EZchip provides two cables: a micro-USB cable and a UART/RS232 debug cable.

Connect your cables

Connect the appropriate Ethernet cables to the installed transceivers on the TILEncore-Gx36. The following cable types might be required:

Table 4-2. Cable Connections

Connection	Cable
SFP w/ Cu cartridge	CAT5e or CAT6 cable. For more information refer to Table 5-4 on page 29.
SFP w/ SX cartridge	1000BASE-SX OM3 Multimode Fiber (typically orange)
SFP+ w/ SR cartridge	10GBASE-SR OM3 Multimode Fiber (typically orange)

Note: Long-range fiber connections can be used, however EZchip does not offer the SFP/SFP+ cartridges for this configuration.

4.4 Jumpers and DIP Switches

This section lists TILEncore-Gx36 jumpers and DIP switch and their default settings.

4.4.1 Jumpers

Table 4-3 lists the TILEncore-Gx36 jumper plug settings and Table 4-4 lists the header settings. Default values are displayed in **boldface** type.

Table 4-3. Jumper Plug Settings

Jumper ID	Reference Designator	Description
Voltage Selector	J8	DDR regulator voltage select. This jumper is used explicitly for the memory interface. Enables users to specify: • 1.5 V with jumper (default) • 1.35 V without jumper

Table 4-4. Connector Definitions

Jumper ID	Reference Designator	Description
UART Header	J1	Connector for UART Cable. Pin 1 Green (TX) Pin 2 Black (GND) Pin 3 Red (RX)
CPLD Programming Header	J2	Xilinx Programming Header.
Sequencer I^2C Access	J3	UCD9081 Power Sequencer Header. Six Pin Header. Pin 1 PRGMR3.3V Pin2 GND Pin 5 SCL Pin 6 SDA
ATX Power.	J4	ATX Power.
USB Connector	J6	Connector for USB 2.0.
Fan Power	J7	Connector for fan heat sink or header for Tile Processor. Pin 1 GND Pin 2 +12V
SFP Port 0	J11	Connector for SFP port 0 ([x]gbe1).
SFP Port 1	J13	Connector for SFP port 1 ([x]gbe2).
SFP Port 2	J15	Connector for SFP port 2 ([x]gbe3).
SFP Port 3	J17	Connector for SFP port 3 ([x]gbe4).
SMA Cable In	J18	SubMiniature version A (SMA) pulse cable input.

4.4.2 DIP Switch (SW2)

The TILEncore-Gx36 card has one DIP switch (SW2), as shown in Figure 4-5 and listed in Table 4-5. See also Figure 2-3 on page 9 for the location of the DIP switches on the bottom of the card.

In Table 4-5 in the "*Description*" section, OFF (or LOW) is equivalent to 0 and ON (or HIGH) is equivalent to 1.

Table 4-5. DIP Switch SW2 Settings

Switch Position	Default Setting	Description
1	**ON**	**CPU_CONFIG_UART_0. UART0 Configuration.** OFF CPU_CONFIG_UART[0]is LOW - Protocol Mode NOT Enabled for UART **ON CPU_CONFIG_UART[0] is HIGH - Protocol Mode Enabled for UART**

Table 4-5. DIP Switch SW2 Settings (continued)

Switch Position	Default Setting	Description
2	**OFF**	**CPU_CONFIG_USB_0. USB0 Configuration.** **OFF USB Endpoint. TLR CONFIG_USB [0] is HIGH - Protocol Mode Enabled for USB** ON USB host. TLR CONFIG_USB [0] is LOW - Protocol Mode NOT Enabled for USB
3	**ON**	**CPU_CONFIG_PCIE0_X8. PCI_X8 Configuration.** OFF x4 lane mode **ON x8 lane mode**
6:4	<u>4</u> <u>5</u> <u>6</u> OFF OFF OFF **ON OFF OFF** OFF ON OFF ON ON OFF OFF OFF ON ON OFF ON OFF ON ON ON ON ON	**PCIe0 Configuration. MODE [2:0], CPU_CONFIG_PCIE0_[0:2].** **<u>Description</u>** 000 DISABLED **001 AUTO CONFIG ENDPOINT** 010 AUTO CONFIG RC 011 AUTO CONFIG ENDPOINT, GEN1 100 AUTO CONFIG RC, GEN1 101 AUTO XLINK 110 STREAM X1 111 STREAM X4
7	**OFF**	**GPIO32 Reserved**. **OFF Software Defined, SOFTWARE_DEF_0 = LOW.** ON Software Defined, SOFTWARE_DEF_0 = HIGH.
8	**OFF**	**GPIO33 SOURCE**. **OFF Always leave in the OFF position.** ON Reads the status of edge connector.
9	**OFF**	**EDGE CONNECTOR 12V ENABLE**. **OFF Edge connector power is enabled.** ON Edge connector power is disabled.
10	**OFF**	**PCIe REFCLOCK SOURCE** **OFF Host provides reference clock.** ON Local reference clock used.

4.5 Installing the TILEncore-Gx36 Drivers

■ **Install drivers:**

For instructions on installing drivers, refer to "*Installing this Release*" in the *Getting Started with the Multicore Development Environment* (UG504).

CHAPTER 5 FUNCTIONAL DESCRIPTION

This chapter describes each of the TILEncore-Gx36™ card's functional subsystems.

5.1 Power

5.1.1 Power Entry

Power is supplied from the ATX power supply connector coupled to the power supply in the chassis or the PCIe edge connector. If your slot is designated as a 75W slot, power can come from the edge connector.

The ATX connector is a standard ATX peripheral power connector that provides +12V, if you elect to use this method of powering your TILEncore-Gx36. All other sub-voltages are generated by switching power supplies or LDO regulators on the board. This arrangement allows for the PCIe card to be put into a “stand-alone” chassis where there are no connections to the edge connector, power or otherwise.

5.1.2 ATX Power Requirements

Only +12V power, with a regulation tolerance equal to +/-5% (11.4V - 12.6V), is required from the ATX connector (5V is no-connect). The power supply must be able to supply 6A.

Note: The PCIe card does not tolerate “sag” or transients outside of these tolerances.

5.1.3 Power Sequencing

On-board power sequencing is controlled by the Texas Instruments UCD9081 voltage monitor/sequencer. The UCD9081 senses when +12V has been applied and enables the switching regulators that supply power for:

- VCore (0.8V – 1.1V)
- 1.0V
- 1.8V
- 3.3V
- DDR3_VDD

The monitor/sequencer (UCD9081) is configured to sequence all these voltages up and down together. When all the voltages are between -6.5% and +25% of their nominal voltage, the power-on reset signal is de-asserted. During operations if one of the power supplies falls outside of these operating ranges, the UCD9081 shuts down the other supplies.

5.1.4 TILE-Gx Over-Temperature Protection

The National Semiconductor LM95235Q temperature sensor has an open-drain output that is pulled low when it reads the TILE-Gx die temperature above a programmed threshold. This Active LOW output travels to two locations:

- GPIO[44] of the TILE-Gx. Software uses this as an early-warning indicator that the card power is going to shut off soon, in case any process clean-up or file-logging needs to be done.
- The UCD9081 power sequencer. An overtemp event puts the sequencer into reset and all voltage rails are disabled.

5.2 Card Resets

The card supports several reset options and each reset source is able to generate a system reset, which asserts a hard reset to the TILE-Gx processor. All other devices are under software control of the Tile processor, so a hard reset of the TILE-Gx processor propagates to all the other resettable devices on the board including the XGbE PHY, USB PHY, and PCIe endpoint when the TILE-Gx is in root complex mode. See Table 5-1 for a reset propagation matrix and the subsections below for further information regarding specific device and interface reset conditions.

Note: All reset lines are active LOW.

Table 5-1. Reset Propagation

Reset Source	TILE-Gx Reset	Description
Power-On	Yes	Controlled by on-board power sequencer/monitor.
PCIe Reset	Yes	When the card is configured as an end point (add-in card), reset is generated by the host. This signal is an output from the card through the gold fingers when the TILE-Gx is configured to operate in Root Complex mode.
Push Button Reset	Yes	Controlled by push button switch (SW1).

5.2.1 Power-On Reset

Power-On reset is controlled by the voltage monitor/sequencer (the Texas Instruments UCD9081), which senses when +12V has been applied and enables the switching regulators. When all the voltages are between -6.5% and +25% of their nominal voltage, the power-on reset signal is de-asserted.

The 12V supply is monitored and sequenced by the Linear Tech LTC4211 Hot Swap controller, which is (by default) enabled by the presence of +12V on the PCIe edge connectors and it can optionally be enabled via +12V present on the ATX connector (via jumper J4). This Hot Swap controller fuses and ramps the +12V power supplied on the ATX connector.

The sections that follow provide information regarding specific device and interface reset conditions.

5.2.2 PCIe Reset

PCIe reset is logically connected to the TILEncore-Gx36 from the PCIe edge connectors and is distributed from an onboard CPLD to the TILE-Gx processor. On reset it causes the TILE-Gx to execute a Level 0 boot, but this depends on whether or not the TILE-Gx is in PCIe root complex mode. When the TILE-Gx is not Root Complex, the PCIe reset is logically connected to the TILE-Gx's hard reset signal and causes the TILE-Gx to execute a Level 0 boot.

Note: There are provisions for the TILE-Gx to act as "Root Complex" in a PCIe system, and generate reset for downstream devices. Contact your FAE for details.

Note: The TILE-Gx has a GPIO pin `GPIO[46]`, which is driven to the Xilinx CPLD. If this signal is asserted (HIGH), then the PCIe reset to the gold fingers is generated by the TILE-Gx via `GPIO[45]`.

5.3 Clock Distribution

There are five clock trees on the PCIe card that are generated from a single clock driver:

- 125 MHz: This clock is distributed, single ended, to the TILE-Gx processor 's:
  - `GX_CORE_REFCLK`
- 100 MHz: This clock is distributed, single ended, to the TILE-Gx processor's:
  - `GX_PCIE_REFCLK`

  —OR—
- 100 MHz: This is a LVPECL clock distribution for PCIe via:
  - `GX_PCIE_REFCLK`

    Note that this clock comes from the edge connector and not the IDT5V49EE904. It is listed here because it can be used instead of the one from the IDT5V49EE904.
- 156.25 MHz: This clock is distributed, single ended, to the TILE-Gx processor's
  - `GX_ENET_REFCLK`.
- 25 MHz: This clock is distributed, single ended, to the CPLD.
- 26 MHz: This clock is distributed, single ended, to the USB3317 USB Controller.
  - `USB_ULPI0_REFCLK`

5.4 TILE-Gx Processor Connections

Table 5-2 provides a list of the Tile Processor interfaces and their connections:

Table 5-2. Tile Processor I/O Connections

Description	Interfaces to...
DDR3 Memory Interface '0'	DDR3 ECC SODIMM
DDR3 Memory Interface '1'	DDR3 ECC SODIMM
XAUI Interface '0'	BCM8747
XAUI Interface '1'	BCM8747
XAUI Interface '2'	BCM8747
XAUI Interface '3'	BCM8747
PCIe Interface '0'	PCIe Edge Connector
PCIe Interface '1'	PCIe Edge Connector

Table 5-2. Tile Processor I/O Connections (continued)

Description	Interfaces to...
PCIe Interface '2'	N/A
GPIO Interface	Various I/Os
UART	On-board RS-232 Transceiver
I^2C Master 0	Boot/BIB ROM
I^2C Master 1	DIMM SPD ROMs
I^2C Master 2	I^2C Peripheral devices (Temp sensors, power monitors, & clock control)
I^2C Slave	N/A
SROM_SPI	On-board SPI ROM
usb0	USB Connector
usb1	Not used.

5.5 I/O Subsystems

5.5.1 SFP+ Connectors for 1G/10G Ethernet

These connectors are located on the board and provide 10 Gbps Ethernet over fiber using standard SFP+ (Small Form Pluggable for 10G) modules. This XAUI SFP+ Connector uses Ports 1-4.

Connector Detail:

- Molex SFP Connector: 20 Circuits Receptacle 30u Au Plating
- Molex P/N 74441-0010

Table 5-3. SFP Connector Summary

Pin	Signal Name	Description[a]
1	VeeT	Module Transmitter ground
2	TxFault	Module Transmitter Fault
3	TxDisable	Transmitter Disable, Turns off transmitter laser output.
4	MOD-DEF(2)	Data - SDA line for the I^2C identification EEPROM
5	MOD-DEF(1)	Clock - SCL line for the I^2C identification EEPROM
6	MOD-DEF(0)	Module Absent - Grounded by the module to indicate that the module is present
7	Rate Select 0 (RS0)	Rate Select 0

Table 5-3. SFP Connector Summary (continued)

Pin	Signal Name	Description[a]
8	RX LOS	Receiver Loss of Signal indication
9	Rate Select 1 (RS1)	Rate Select 1
10	VeeR	Module Receiver Ground
11	VeeR	Module Receiver Ground
12	RD-	Receiver Inverted Data Output
13	RD+	Receiver Non-Inverted Data Output
14	VeeR	Module Receiver Ground
15	VccR	Module Receiver 3.3V Supply
16	VccT	Module Transmitter 3.3V Supply
17	VeeT	Module Transmitter Ground
18	TD+	Transmitter Inverted Data Output
19	TD-	Transmitter Non-Inverted Data Output
20	VeeT	Module Transmitter Ground

a.Per MSA SFF-8431 Specification.

Table 5-4 lists EZchip-qualified SFP optical modules.

Note: Use only Certified Class 1 lasers or Certified SFPs.

Table 5-4. Qualified SFP Optical and Copper Modules

SFP Modules	Make	Model Number
SFP Optical Modules		
1GbE	Avago	AFBR-5715APZ
	Finisar	FTLF8519P2BCL
	Agilent	HFBR 5710L
SFP Copper Modules		
1 Gigabit 1000BASE-T non-RoHS	Finisar	FCMJ-8521-3
1 Gigabit 1000BASE-T RoHS	Finisar	FCLF-8521-3

Table 5-4. Qualified SFP Optical and Copper Modules (continued)

SFP Modules	Make	Model Number
1.25 GBd Electrical Transceiver for 1000BASE-T, SFP, RX_LOS Disabled, RoHS-Compliant	Avago	ABCU-5710RZ

Table 5-5 lists EZchip-qualified SFP+ optical modules.

Note: Use only Certified Class 1 lasers or Certified SFPs.

Table 5-5. Qualified SFP+ Optical Modules

SFP+ Modules	Make	Model Number
SFP+ Optical Modules		
10Gb	Avago	AFBR-703SDZ
	Finisar	FTLX8571D3BCL

5.5.2 UART Main Board Connector

Connector Detail:

- Reference designator J1 (UART header)
- Tyco connector
- Three circuit, .25 square pins straight post, RoHS-compliant
- Tyco P/N: 3-644456-3

Note: The UART connection is fully accessible in all configurations of the TILEncore-Gx36. Refer to Figure 2-2: TILEncore-Gx36 with SFP/SFP+ Interfaces on page 8 for an illustration of the TILEncore-Gx36.

Table 5-6. RS232 Connector Summary

Pin	Serial Cable Wire Color	Signal Name	DB-9 Female Serial RS232 Pin Number	Description
1	Green	RS232_TX	J1 pin 1 is DB-9 pin 2	TILE-GX RS232 Data Output
2	Black	GND	J1 pin 2 is DB-9 pin 5	Power Return
3	Red	RS232_RX	J1 pin 3 is DB-9 pin 3	TILE-GX RS232 Data Input

5.6 Miscellaneous Connections

5.6.1 Fan Power Connector

Connector Detail:

- Reference designator J7
- Molex connector
- CONN, HEADER, 2P, 1.25MM, VERT, SMT
- Molex P/N: 53398-0271

Table 5-7. Fan Power Connector Summary

Pin	Signal Name	Description
1	Ground	Power Return
2	+12V	Fan Power

5.6.2 ATX Power Connector

The ATX Power connector is used to supply +12V to the TILEncore-Gx36 board. The TILEncore-Gx36 board does not utilize any of the power pins from the PCIe Edge Connector.

Connector detail:

- Reference designator J4
- Molex connector
- Header, six circuits, right angle mount, through hole, RoHS-compliant
- Molex P/N: 0457320001

Table 5-8. ATX Power Connector Summary

Pin	Signal Name	Description
1	+12V	12V Power
2	+12V	12V Power
3	+12V	12V Power
4	Sense	Sense Tied to Ground
5	Ground	Power Return
6	Ground	Power Return

5.7 DDR3 Memory Configuration Subsystem

The two memory interfaces of the TILE-Gx are each connected to a DIMM socket for easy user modification of speed and density. Though we have a target configuration of 4Gbyte/interface density, component availability or customer requirements make these interfaces a “build option”. Configuration data for these memory interfaces is stored in the DIMM SPD (2-wire serial) ROM so that the TILE-Gx software can configure its memory controllers to work correctly with the memory.

Note: Only ECC SODIMMs should be used in the TILEncore-Gx36.

5.7.1 SODIMM Connectors

SODIMM Connector Details

Tyco Connector:

- Connector, 204 conductors, 5.2mm Height, reverse orientation, RoHS-compliant
- Tyco P/N: 2-2013290-3

Tall Connector Detail

Tyco Connector:

- Connector, 204 conductors, 9.2mm Height, Standard orientation, RoHS-compliant
- Tyco P/N: 2-2013310-3

SODIMM Connector Assignments

DIMM connector numbers include the following:

- SODIMM MEM0 J5
- SODIMM MEM1 J9

5.8 TILE-Gx36 Processor’s GPIO

Table 5-9 details the connections to the GPIO pins of the Tile Processor.

Table 5-9. TILE-Gx Processor’s GPIO Assignment

Pin Name	Usage Class	Usage Detail
GPIO[0]	RFU	SMA Pulse
GPIO[1]	RFU	Reserved For Future Use
GPIO[2]	RFU	Quad PHY Reset (level translated via the CPLD)
GPIO[3]	RFU	Reserved For Future Use
GPIO[4]	RFU	Reserved For Future Use
GPIO[5]	RFU	Reserved For Future Use
GPIO[6]	RFU	Reserved For Future Use
GPIO[7]	RFU	Reserved For Future Use

Table 5-9. TILE-Gx Processor’s GPIO Assignment (continued)

Pin Name	Usage Class	Usage Detail
GPIO[8]	RFU	Reserved For Future Use
GPIO[9]	RFU	Reserved For Future Use
GPIO[10]	RFU	Reserved For Future Use
GPIO[11]	RFU	Reserved For Future Use
GPIO[12]	RFU	Reserved For Future Use
GPIO[13]	RFU	Reserved For Future Use
GPIO[14]	RFU	Reserved For Future Use
GPIO[15]	RFU	Reserved For Future Use
GPIO[16]	RFU	Reserved For Future Use
GPIO[17]	RFU	Reserved For Future Use
GPIO[18]	RFU	Reserved For Future Use
GPIO[19]	RFU	Reserved For Future Use
GPIO[20]	RFU	Reserved For Future Use
GPIO[21]	RFU	Reserved For Future Use
GPIO[22]	RFU	Reserved For Future Use
GPIO[23]	RFU	Reserved For Future Use
GPIO[24]	RFU	Reserved For Future Use
GPIO[25]	RFU	Reserved For Future Use
GPIO[26]	RFU	Reserved For Future Use
GPIO[27]	RFU	Reserved For Future Use
GPIO[28]	RFU	Reserved For Future Use
GPIO[29]	RFU	Reserved For Future Use
GPIO[30]	RFU	Reserved For Future Use
GPIO[31]	RFU	Reserved For Future Use
GPIO[32]	SOFTWARE_DEF_0	Software Defined dip switch jumper
GPIO[33]	EDGE_PWR_DISABLE_L	The processor monitors the state of SW2 position 9.

Table 5-9. TILE-Gx Processor's GPIO Assignment (continued)

Pin Name	Usage Class	Usage Detail
GPIO[34]	RFU	Unused, read as 0.
GPIO[35]	PCIE_CLOCK_SEL_PU	Monitor DIP switch that selects between edge connectors or local PCIe clock
GPIO[36]	RED_LED_FAIL	Drives RED Fail LED
GPIO[37]	USB_OC_FLAG_L	USB Over Current FLAG_L, input, active low
GPIO[38]	RFU	Reserved For Future Use
GPIO[39]	RFU	Reserved For Future Use
GPIO[40]	GX_USB_RESET_L	Reset to CPLD for USB Controller, output, active low.
GPIO[41]	RFU	Reserved For Future Use
GPIO[42]	RFU	Reserved For Future Use
GPIO[43]	GX_POWER_INT_L	From Xilinx CPLD, input, active low.
GPIO[44]	GX_DIE_TEMP_INT_L	TILE-GX Die Temperature Interrupt, input, active low
GPIO[45]	GX_PCIE_PERST_N	To Xilinx CPLD, output
GPIO[46]	PCIE_ROOT_COMPLEX_SEL	To Xilinx CPLD, output
GPIO[47]	GX_XAUI0_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[48]	GX_XAUI0_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[49]	GX_XAUI0_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[50]	GX_XAUI0_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[51]	GX_XAUI1_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[52]	GX_XAUI1_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[53]	GX_XAUI1_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[54]	GX_XAUI1_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[55]	GX_XAUI1_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[56]	GX_XAUI2_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[57]	GX_XAUI2_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[58]	GX_XAUI2_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[59]	GX_XAUI2_LASI	XAUI PHY Link Alarm Status Interrupt

Table 5-9. TILE-Gx Processor's GPIO Assignment (continued)

Pin Name	Usage Class	Usage Detail
GPIO[60]	GX_XAUI3_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[61]	GX_XAUI3_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[62]	GX_XAUI3_LASI	XAUI PHY Link Alarm Status Interrupt
GPIO[63]	GX_XAUI3_LASI	XAUI PHY Link Alarm Status Interrupt

5.8.1 TILE-Gx Processor's I^{2}C — Master Interface

There are currently three master interfaces connected to the TILE-Gx Processor's I^{2}C Master interface directly mounted on the TILEncore-Gx36 card. There are currently 15 devices connected to the TILE-Gx's I^{2}C Master Interface directly mounted on the TILEncore-Gx36 card. Table 5-10 lists the device address, part type, and description.

Note: While there is one slave interface on the TILE-Gx processors' I^{2}C slave interface, the slave interface is unused on the TILEncore-Gx36 design.

Table 5-10. I^{2}C Address and Device Map

I^{2}C Address 7-bit Address	Vendor and Part Number	Description
1010 100X	Atmel - AT24C512	2-wire ROM, 512 Kbits (Atmel)
1010 000X	Atmel - type ROM	DDR3 I/F 0 SPD ROM - DIMM
1010 001X	Atmel - type ROM	DDR3 I/F 1 SPD ROM - DIMM
1110 000X	Texas Instruments PCA9546A	4 Channel I2C Switch.
1100 000X	Texas Instruments UCD9081	8 Channel Power Sequencer and Monitor
1000 000X	Texas Instruments INA219	Current/Power Monitor (VCORE VRM)
1000 010X	Texas Instruments INA219	Current/Power Monitor (MEM0 VDDR3 VRM)
1000 011X	Texas Instruments INA219	Current/Power Monitor (MEM1 VDDR3 VRM)
1000 110X	Texas Instruments INA219	Current/Power Monitor (GX36 1.8V)
1000 100X	Texas Instruments INA219	Current/Power Monitor (GX36 ENET 1.0V)
0100 010X	Intersil ZL2106	3.3V Power Supply
0100 000X	Intersil ZL2106	1.8V Power Supply
0100 001X	Intersil ZL9101M	1.0V Power Supply
0011 000X	National Semi LM95235QQ	TILE-Gx36 Temperature Sensor

Table 5-10. I^2C Address and Device Map (continued)

I^2C Address 7-bit Address	Vendor and Part Number	Description
1101 010X	IDT IDT5V49EE904	Clock Driver Chip

5.8.1.1 I^2C ROM

An I^2C 2-wire serial EEPROM is available for user configuration data. It is connected to the TILE-Gx Processor's I2CM0 interface. The device is an Atmel AT24512 or equivalent 512K part.

5.8.1.2 I^2C Temperature Sensor

The National Semiconductor LM95235Q Temperature sensor is used to monitor the temperature of the TILE-Gx Processor to prevent damage to the device due to overheating. The LM95235Q senses its own temperature as well as that of the diode junction resident on the Tile Processor's die.

The LM95235Q has an open-drain output that is pulled LOW when it reads the TILE-Gx Processor die temperature above a programmed threshold. This active low output goes to two places: The TILE-Gx Processor and CPLD.

- Tile Processor

  `GPIO00[44]` of the TILE-Gx. This pin could be used by software as an early warning indicator that the board power is soon to be shut off, in case there is any process clean-up or file-logging that needs to be done prior to the UCD9081 power controller beginning the power down sequence.

5.8.2 TILE-Gx Processor 4-Wire Interface

5.8.2.1 TILE-Gx Processor SPI FLASH Memory

There is a SPI ROM for boot loading or other user configuration data. It is connected to the TILE-Gx's SPI interface. The device is a Numonyx N25Q128, 128Mbit part.

5.8.3 TILE-Gx External Interrupts

Table 5-11 details the interrupts to the TILE-Gx.

Table 5-11. TILE-Gx Processor's Interrupt Assignment

Interrupt Name	Interrupt Source	Physical Connection
GX_POWER_INT_L	Xilinx CPLD	TILE-Gx: GPIO00[43]
GX_DIE_TEMP_INT_L	LM95235Q Temperature sensor - When asserted LOE provides an early warning that board power is going down due to over temperature.	TILE-Gx: GPIO00[44]
XAUIx_LASI	BCM8747	TILE-Gx: GPIO00[48:63]

5.9 LEDs

5.9.1 Power and Status Indicators

The main board contains four LEDs that include three power-related LEDs and one software-programmable red LED on the card, as listed on Table 5-12. Figure 5-1 shows the location of the Power On and Power OK LEDs with the board in standalone mode. Note that this does not include Port Link Status/Activity LEDs that are controlled by the PHY (see Figure 7-1).

For more information on Link Status LEDs, refer to Figure 2-2: TILEncore-Gx36 with SFP/SFP+ Interfaces on page 8 or Figure 7-1: SFP/SFP+ Link and Power LEDs on page 45.

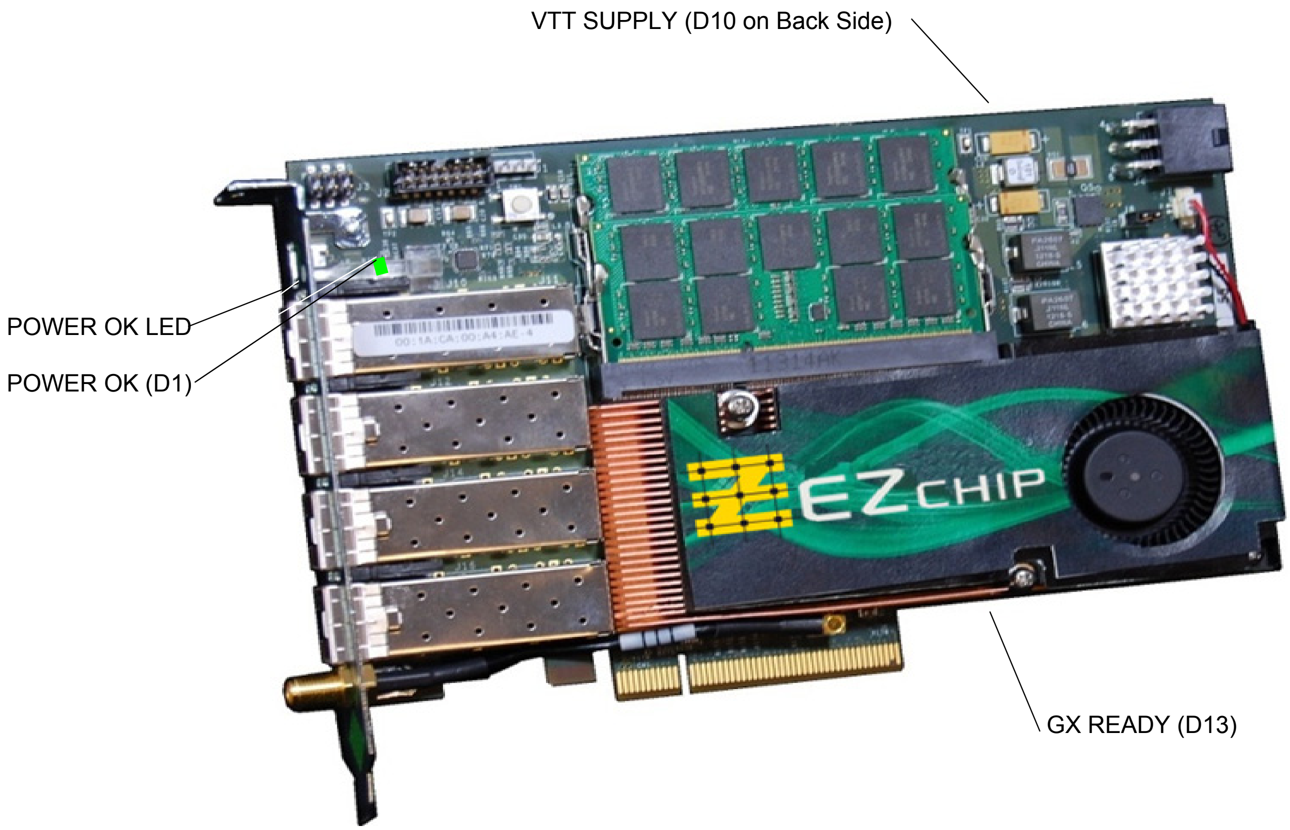

Figure 5-1: TILEncore-Gx Board LEDs— Top Side

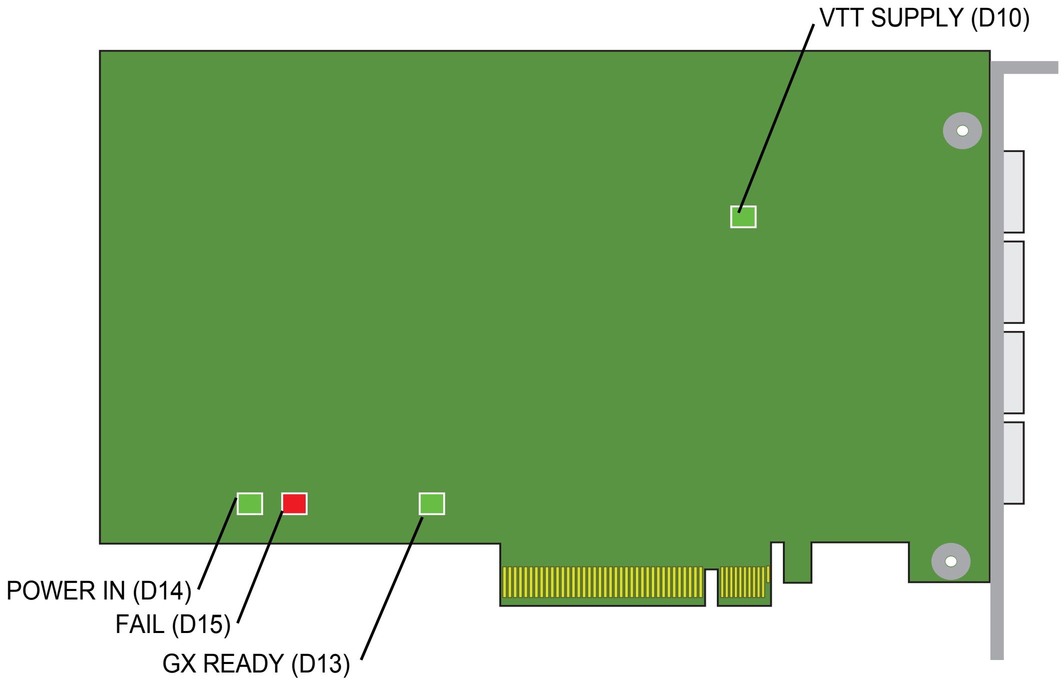

Figure 5-2: TILEncore-Gx Board LEDs— Bottom Side

Table 5-12. Board LED Definitions

Reference	Color	Label	Location	Description
D14	GREEN	POWER IN	Back	The hot swap controller is enabled and +12 V is being provided to the card's on-board regulators.
D1	GREEN	POWER OK	Front	This LED is driven from the regulated 3.3 V to indicate that all secondary voltages are up and running on the card.
D13	GREEN	GX READY	Back	This LED indicates that the core RefClk is present, reset has been removed, and the internal configuration is complete.
D15	RED	FAIL	Back	Driven by software via a GPIO[36] on the TLR4-03680. This LED is OFF by default.
D10	GREEN	VTT SUPPLY	Back	The VTT Regulator is operating correctly.

5.10 Power LED

The Power LED, located on the Rear Plate glows green only when all components are powered on (see Figure 5-3).

Figure 5-3: SFP/SFP+ Link and Power LEDs

5.10.1 Interface Status Indicators

Status indicators on the rear plate show the state of the Ethernet links.

A pair of LED indicators is located next to each port. The green indicator is for link status (up or down) and the yellow indicator is for activity.

CHAPTER 6 RUNNING PROGRAMS ON THE TILENCORE-GX36

6.1 Problems with Simultaneous USB and PCIe Usage

The TILEncore-Gx™ card supports boot and debug operations via its PCIe interface, as well as via its USB device interface. It is possible to connect both of these interfaces simultaneously, but doing so can cause undesired system behavior. This section explains why that is and how to avoid it.

The problem is related to booting the TILE-Gx™ processor, which happens whenever you run `tile-monitor` *without* the `--resume` option, or whenever you run any chip tests via the Board Test Kit (BTK). The first step of booting the TILE-Gx processor is to reset it. This operation is done automatically by the `tileusb` or `tilegxpci` drivers when you start to boot the processor via the corresponding interfaces.

When the processor is reset, all of the I/O interfaces on the chip get reset. In the case of USB, the USB device appears to be disconnected from the host; in the case of PCIe, the PCIe link between the host and the processor goes down. Soon afterward, communication is re-established.

In the case of USB, the link is brought back up automatically by the processor; in the case of PCIe, the PCIe prebooter is loaded from the SROM and brings up the link. However, the host's response to these momentary interruptions is very different. USB is designed as a hot-pluggable interface. The cabling for USB devices is generally accessible to their users, and many of those devices are connected to their host system only temporarily (as when you download photographs from a digital camera). Because of this, the USB software stack expects devices to be disconnected and reconnected.

While PCIe devices can also be hot-pluggable, this is the exception, rather than the rule. Desktop computers, which are commonly used to host TILEncore-Gx36 cards, typically do not support hot-plugging. Thus, their PCIe software stacks do not expect PCIe devices to be disconnected.

In order to cope with the fact that rebooting (and thus resetting) a TILE-Gx processor causes the USB or PCIe connection to be interrupted, the MDE's `tileusb` and `tilegxpci` drivers perform special steps as part of a reboot operation. The driver might need to read the configuration state from the device, though state information is destroyed by the reset. Once the device has been reconnected, the configuration state is restored. The driver might also need to reconfigure the system temporarily to tolerate the upcoming disconnect event. However, these steps can only be performed if the relevant driver knows that a reset is about to happen, which brings us to the problem with simultaneous USB and PCIe connections.

Imagine what happens when a TILEncore-Gx card is configured into a host system and both its USB and PCIe ports are connected. Both the `tileusb` and `tilegxpci` drivers believe they are responsible for controlling the card, and both devices are willing to reboot and thus reset it. However, neither driver has any "knowledge" of the other. Thus, if you use the `tileusb` driver to reset the card, the `tilegxpci` driver does not get the opportunity to do any of the normal state-saving and disconnection preparations; the host system sees an unexpected interruption of service to one of its PCIe devices instead.

The effect of this kind of event is system-dependent. On some systems, the unexpected PCIe disconnect leads to a fatal error and a system crash. On others, the system does not crash, but the fact that the newly-reset TILEncore card has lost the configuration information programmed into it during device discovery can lead to problems with PCIe I/O, perhaps not even involving the TILEncore card. In either case, the only way to recover is to reboot the host system.

Because of this, we strongly recommend that if you have a TILEncore-Gx36 connected via both PCIe and USB, that you never reboot the card via USB.

This means that you should not run `tile-monitor` against the USB interface, and should not run any tests when using `tile-btk` against the USB interface. Rebooting such a doubly-connected card via PCIe, on the other hand, is generally safe, since the USB interface is much more tolerant of momentary interruptions.

If you have PCIe and USB connected at the same time, **you should avoid resetting the card via USB**.

If you reset the card over USB, one of two things is going to happen:

- If your card is in a host **that is not sensitive to the PCI link going down**, the PCI link goes down and comes back up, but the chip is not programmed properly and you are be able to talk to the card over PCIe. It is also possible that the default settings on the card might interfere with other things on the BUS, which is less likely but still possible. The main symptom is that you are not able to talk to the card over PCIe. The only way to fix that problem is to reboot the host system.
- If your card is in a host **that is sensitive to the link going down**, the system most likely crashes and reboots. Some hosts might not be set up to reboot after a failure like that. You might need to power cycle the host in these cases.

6.2 Boot Options

The TILEncore-Gx36 card supports the following boot options:

- Boot from host via the TILEncore-Gx36™ card's PCIe 0 or 1 Interface.

  This is the standard boot option.
- Boot from boot ROM (SPI ROM).

  Requires programming of SPI ROM.
- Boot from UART.

  UART port configured in `UART_PROTOCOL_MODE`.
- Boot from USB.

Refer to the DIP switch configuration information in Section 4.4 Jumpers and DIP Switches for settings related to boot options.

6.2.1 Booting a TILEncore-Gx36 via USB

To run a program on the TILEncore-Gx36™ card for exhibition purposes, use the `--dev` option:

```
tile-monitor --dev usb0 \
--hvd POST=quick \
--hvx TLR_NETWORK=none \
--hvx "dataplane=1-35" --hvd CPU_SPEED=1200 \
```

This automatically reboots the card. If you only have one card in your system, you can leave off the `--dev usb0`, for example:

```
tile-monitor -- <executable> [<argument> ...]
```

You can also run against an already-booted TILEncore-Gx36 card without rebooting it by using the `--resume` option:

```
tile-monitor --dev usb0 --resume
```

6.2.2 Booting a TILEncore-Gx36 via a PCI

```
tile-monitor --dev gxpci0 \
--hvx TLR_NETWORK=none --hvd POST=quick \
--hvx "dataplane=1-35" --hvd CPU_SPEED=1200 \
--hvd CHIP_WIDTH=6 --hvd CHIP_HEIGHT=6 \
--vmlinux $TILERA_ROOT/tile/boot/vmlinux \
--hvc $TILERA_ROOT/tile/etc/hvc/vmlinux.hvc \
```

The `--vmlinux` argument is appended with the default linux kernel file, but shows how you could include some other built kernel file(s).

The `--hvc` argument is appended with the default hypervisor configuration file, but shows how you could include a modified `.hvc` file. Refer to *MDE System Programmer's Guide* for more information on how to configure an alternative hypervisor config then configure and build an alternate vmlinux kernel.

6.3 Specifying Linux Device Paths

You can specify the Linux device paths associated with the TILEncore-Gx36. You only do this if, for example, you have installed the board in a non-standard location:

```
tile-monitor --dev pci0
 [--pci-boot   <boot-device-path>]
```

The `--usb-boot` argument is the device path used to boot the TILEncore-Gx36. This defaults to:

`/dev/tilegxpciN/boot` (in format)

`/dev/tilegxpci0/boot` (default)

Note: If you specify `--resume` you do not need to specify `--usb-boot`, because `tile-monitor` assumes the board is already booted.

Chapter 7 Verifying that the Card Operates Properly

7.1 Power and Status LEDs

The TILEncore-Gx36™ card has four Power and Health LED indicators located on top and bottom of the card. These are defined in Table 5-12 on page 38. When a card is in good working order, powered up, and ready for operation, all four GREEN LEDs (D14, D1, D13, and D10) should be lit. The RED LED (D15) should NOT be lit.

For additional information, refer to Section 5.9 *"LEDs"* on page 37.

7.2 Power LED

The Power LED, located on the Rear Plate glows green only when all components are powered on (see Figure 7-1).

7.3 Port Link Status/Activity LEDs

The LINK LEDs, located on the bracket and depicted in Figure 7-1, indicate that the XAUI links are operating properly.

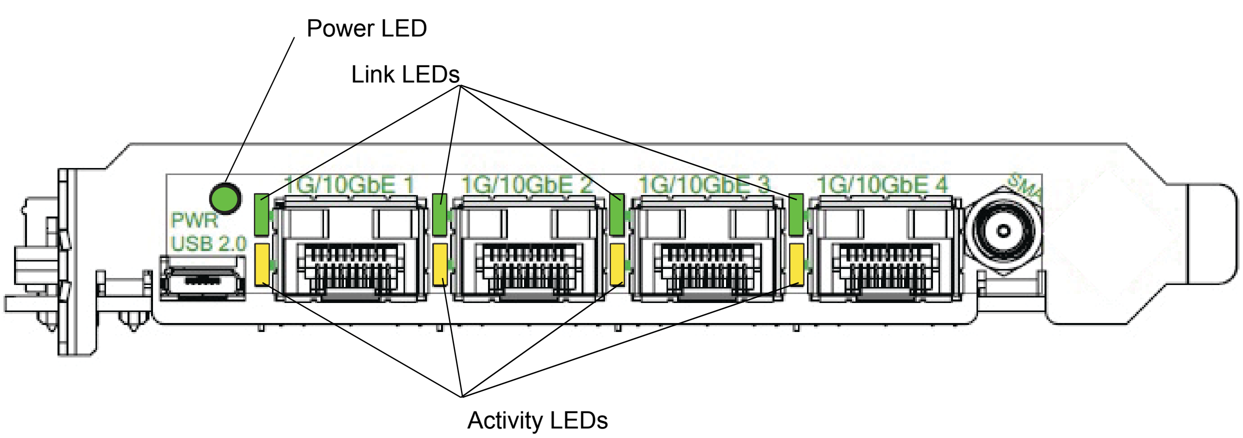

Figure 7-1: SFP/SFP+ Link and Power LEDs

To test the TILEncore-Gx36:

The quick test, `tile-btk`, can be run to test the chip's internal data paths, basic peripherals, XAUI Internal loopback, and memory. Run `tile-btk` from a terminal window at the `btk` prompt.

Note: The user interface of the test kit should be invoked as follows:

```
$TILERA_ROOT/bin/tile-btk
```

For more information about using the Board Test Kit (BTK), refer to *Board Test Kit Guide for TILE-Gx Processors* (UG422).

APPENDIX A: SPECIFICATIONS

A.1 Electrical Specifications

Power consumption varies depending on application, but Table A-1 can be used as a guide:

Table A-1. Power Consumption

Card Version	Typical	Maximum
TILEncore-Gx36 card @ 1.0 GHz	50W (4.2A @ 12V)	60W
TILEncore-Gx36 card @ 1.2 GHz	60W (5A @ 12V)	72W

See also Section 3.2 *"Power Requirements"* on page 12.

A.2 Mechanical Specifications

The mechanical dimensions of the card are compliant with the add-in card form factor for an 8-lane PCI Express "short" add-in card.

Table A-2. Physical Specifications

Board Height	11.1 cm (4.38 in)
Board Length	16.8 cm (6.6 in)
Bracket Height	12 cm (4.72 in)

A.3 Shock and Vibration Specifications

The TILEncore-Gx36 is built to withstand shock and vibration that would be typical in shipping and in non-mobile operating environments. If the board is to be shipped while installed in a host server, it is recommended that the rear bracket be firmly secured into the chassis and that the opposite edge of the card (with the DC power connector) be well supported.

Table A-3 lists the shock and vibration test specifications TILEncore-Gx36 card.

Table A-3. Table A-3. Shock and Vibration Specifications

Specification	Range
Random Vibration	5 - 500 Hz @ 1G rms (9.81 m/s^2 rms), X, Y, and Z axis, 15 min each axis
Shock	Half-Sine @ 80G (785 m/s^2), 2 msec duration, +/- X, Y, Z axis

Note: Detailed test reports can be provided upon request.

A.4 Environmental Specifications

A.4.1 Environment

Table A-4 lists the other environmental specifications of the TILEncore-Gx36 card.

Table A-4. Chassis-Level Environmental Specifications

Specification	Range
Storage temperature	-40° C to +70° C
Operating temperature	0° C to +40° C 1.2 GHz This represents ambient temperatures surrounding the TILEncore-Gx36 card.
Operating humidity	5% to 95%, non-condensing
Air Flow	Nominal chassis airflow required to maintain ambient temperature limits around board is as follows: • 100 lfm @ 40°C with no adjacent slot card • 400 lfm @ 40°C with adjacent slot card

Appendix B: Safety and Regulatory Compliance

B.1 Introduction

This chapter explains important safety precautions that you should observe when operating the TILEncore-Gx36™ card.

Note: Follow all warnings and instructions marked on the equipment.

B.2 Safety Precautions

B.2.1 Use Proper Grounding

To avoid electric shock, make sure that the system is connected to an electrical safety (green wire) ground.

B.2.2 Use ESD Protections

The TILEncore-Gx36 contains static-sensitive devices. Static control procedures should be used when installing, removing, and handling all components. Use an antistatic wrist strap at all times. Wrist straps usually come with an alligator clip that you clip to the metal frame of the chassis after the cover is removed.

Never touch the Card with any conductive objects.

For additional handling instructions, refer to Section 4.1.2 *"Handling the TILEncore-Gx36 Card"* on page 15.

B.2.3 Use Caution When Working Inside a Chassis

Use caution when working inside the computer chassis as touching or bumping the card's rotating blower module can damage it.

B.2.4 Avoid Fumes and Explosives

Do not operate the TILEncore-Gx36 card in the presence of explosives or inflammable gases or fumes.

B.3 Regulatory Compliance

B.3.1 Emissions and Susceptibility Certifications

TBD

B.3.2 FCC Notice

This equipment generates, uses, and can radiate radio frequency energy. If the Card is not installed correctly and used in accordance with the instruction manual, it can cause interference to radio communications.

B.3.3 RoHS Statement

The TILEncore-Gx36 is built to comply with the 2002/95/EC European legislation of 2003-02-13, which states that equipment must be free of:

- Lead
- Mercury
- Cadmium
- Hexavalent Chromium
- Polybrominated Biphenyls (PBB)
- Polybrominated Diphenyl Ether (PBDE)

Products free of the materials is known as being RoHS-6 compliant. However, the TILE-Gx process has a small amount of lead contained in the internal die bond for the “flip chip” package. This means that at the board level, this product is specified to be RoHS-5 compliant. However, there are no other devices containing lead on the PCB, and the soldering technology for all components is lead-free.

Appendix C: Troubleshooting

This chapter describes basic methods for troubleshooting your TILEncore-Gx36™ card.

How do I install the PCIe prebooter?

Boot the board using USB as described in Section 6.2 *"Boot Options"* on page 42. Connect a serial console device to the UART.

From the console port, type:

```
sbim --info
sbim --install-prebooter /boot/pci-preboot.bin
```

At the prompt, respond with "`Yes`" and then enter:

```
sbim --info
```

The system responds with something like:

```
SROM booter: Present 8336 bytes
    Time Installed: Tue
```

What does `tile-console` do?

The `tile-console` command provides access to the console port on a Tile Processor, which displays Linux boot messages and normally provides an interactive Linux login, via a serial device on the host.

Use the same device name nomenclature as with `tile-monitor.` For example:

`--dev usb0`

—OR—

`--dev gxpci0` (default)

With tile-btk operations, I cannot reboot the TILEncore-Gx36 card with -- dev gxpci0 after boot it with --dev usb0. Is this expected?

Yes, you can expect this behavior. We recommend power-cycling the host *after* every time you run the btk commands.

Even if the btk can reboot the TILE-Gx from the SROM and potentially get the PCIe link up, the host does not assign any BAR space to the TILE-Gx, unless we support PCI re-scan, which we do not. Therefore, no further PCI communication can happen.

Appendix D: Glossary

Term	Definition
2-WIRE	2-Wire Serial Clock and Data (SCL and SDA) operating up to 400k Hz.
4-WIRE	4-WIRE, or two pair, Synchronous Serial Interface or Serial Peripheral Interface (SPI).
BIB	Board Information Block. The EZchip® run-time software stack uses a set of system-specific information to govern the configuration and initialization of the Tile Processor™ and the devices to which it connects. This data is contained in a structure called the Board Information Block (BIB).
CPLD	Complex Programmable Logic Device
DDR3	Double Data Rate, Type 3. Synchronous dynamic random access memory—a modern kind of dynamic random access memory (DRAM) with a high bandwidth interface.
DIP	Dual In-line Package.
DIMM	Dual In-line Memory Module • LRDIMM – Load reduced • RDIMM – Registered • UDIMM - Unbuffered
ECC SODIMM	Error Check and Correction Small Outline DIMM.
EEPROM	Electronically Erasable Programmable Read-Only Memory.
FRU	Field Replaceable Unit.
Gbps	Gigabit per second.
GBps	Gigabytes per second.
GbE	Gigabit Ethernet.
GPIO	General Purpose I/O.
I/O	Input/Output.
JTAG	Joint Test Action Group
LFM	Linear Feet per Minute
LED	Light-Emitting-Diode used for status indicator
LVPECL	Low-Voltage Positive-referenced Emitter-Coupled Logic
MDC/MDIO	Management Clock and Data I/O Interface
MDE™	Multicore Development Environment™. EZchip's multicore programming environment.

Term	Definition
PCI	Peripheral Component Interconnect, a bus standard for connecting peripheral devices to a computer motherboard.
PCIe or PCI-e	PCI Express, a PCI interface format using high-speed serial links for the connection between peripheral devices and the motherboard.
POL	Point Of Load.
POR	Power-On Reset.
ROM	Read-Only Memory.
SERDES	Serializer/Deserializer.
SODIMM	Small Outline DIMM.
SPD ROM	Serial Presence Detect ROM.
SPI	Serial Peripheral Interface.
SPI-SROM	Serial Flash with serial peripheral interface.
UART	(Universal Asynchronous Receiver Transmitter). The electronic circuit that makes up the serial port. Also known as “universal serial asynchronous receiver transmitter” (USART), it converts parallel bytes from the CPU into serial bits for transmission, and vice versa. It generates and strips the start and stop bits appended to each character.
USB	Universal Serial Bus
VID	Voltage Identification (bits)
XAUI	10 Gigabit Attachment Unit Interface, a standard for connecting 10 Gigabit Ethernet.
XGbE	10 Gigabit Ethernet

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